

AGENDA

- What is Embedded System ?
- Need for Serial Communication Protocols in ES.
- Intra Circuit vs Inter Circuit Communication ?
- What is Network ?
- What is a Protocol ?
- Low Level /BUS Protocol vs High Level Protocol.
- Bit-Banging Method vs Peripheral Method of Bus Protocol Implementation ?
- Embedded Networking.
- Introduction to UART's protocol.

- What is Network ?

Network refers to the interconnection of two or more devices wired/wireless, directly/indirectly to accomplish reliable communication using one or more protocols functioning between them.

- What is protocol ?

Refers to a set of rules, regulations , checkpoints & sometimes standards to be followed for reliable communication.

- **Low Level protocols** involves rules, regulations, & checkpoints (logical aspect) & standards (physical aspect) as well.

Examples:

Wired protocols (Bus Protocols):

UART's, I2C, SPI, 1-Wire, CAN, LIN, Modbus, USB, etc

Wireless protocols:

(RC5, RC6, Sony, etc)-Remote Control
, Bluetooth, RF, WIFI, NFC, etc

- **High level protocols :**

Involves ONLY rules , regulation & checkpoints (logical aspect).

BUT Standards (Physical Aspects) ARE NOT A Concern.

Hence Depend On LOW LEVEL Protocols to Convey Information Across Physical Wired/Wireless Medium.

- **What is protocol suite ?**

Refers to set/collection/bunch/stack/layers of protocols (BUT NOT just ONE protocol) being used for communication between connected devices.

Examples:

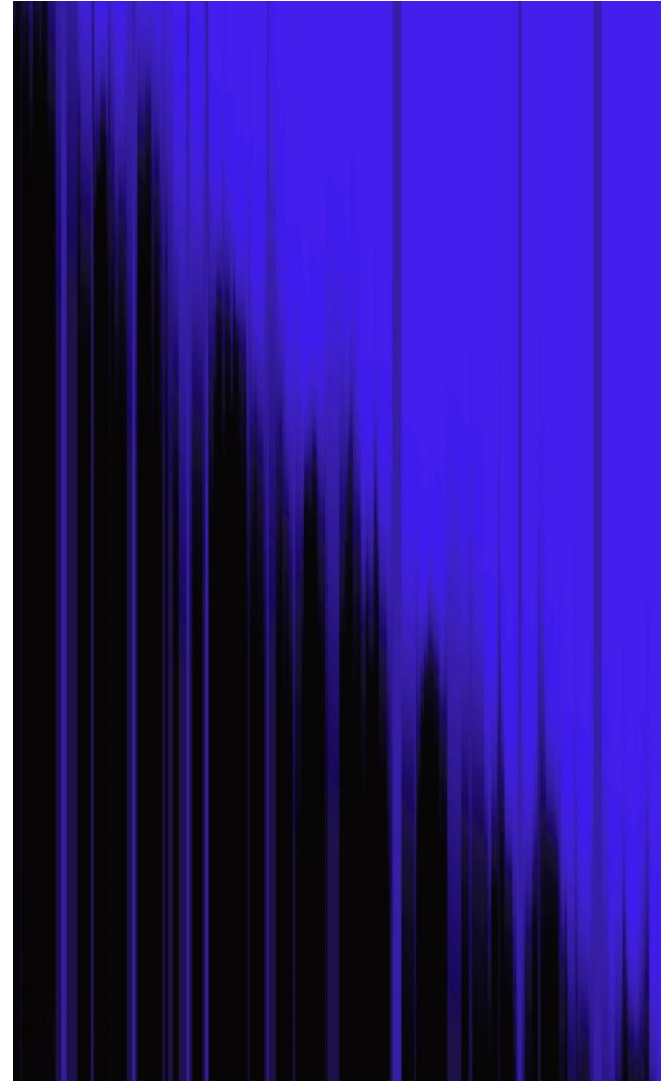
TCP/IP, AUTOSAR/CANOpen, MilCAN, CANAerospace, RJ1939, CoAP, etc

- **Embedded Networking**

Communication ,between two or more embedded devices(inter-circuit) ,or between 2 or more CPUs(intra circuit),or between CPU(s) & certain peripheral components on the same embedded board(intra circuit) can be referred to as embedded networking.

Wired & wireless protocols may be involved.

UART's Communication Protocol



RFID

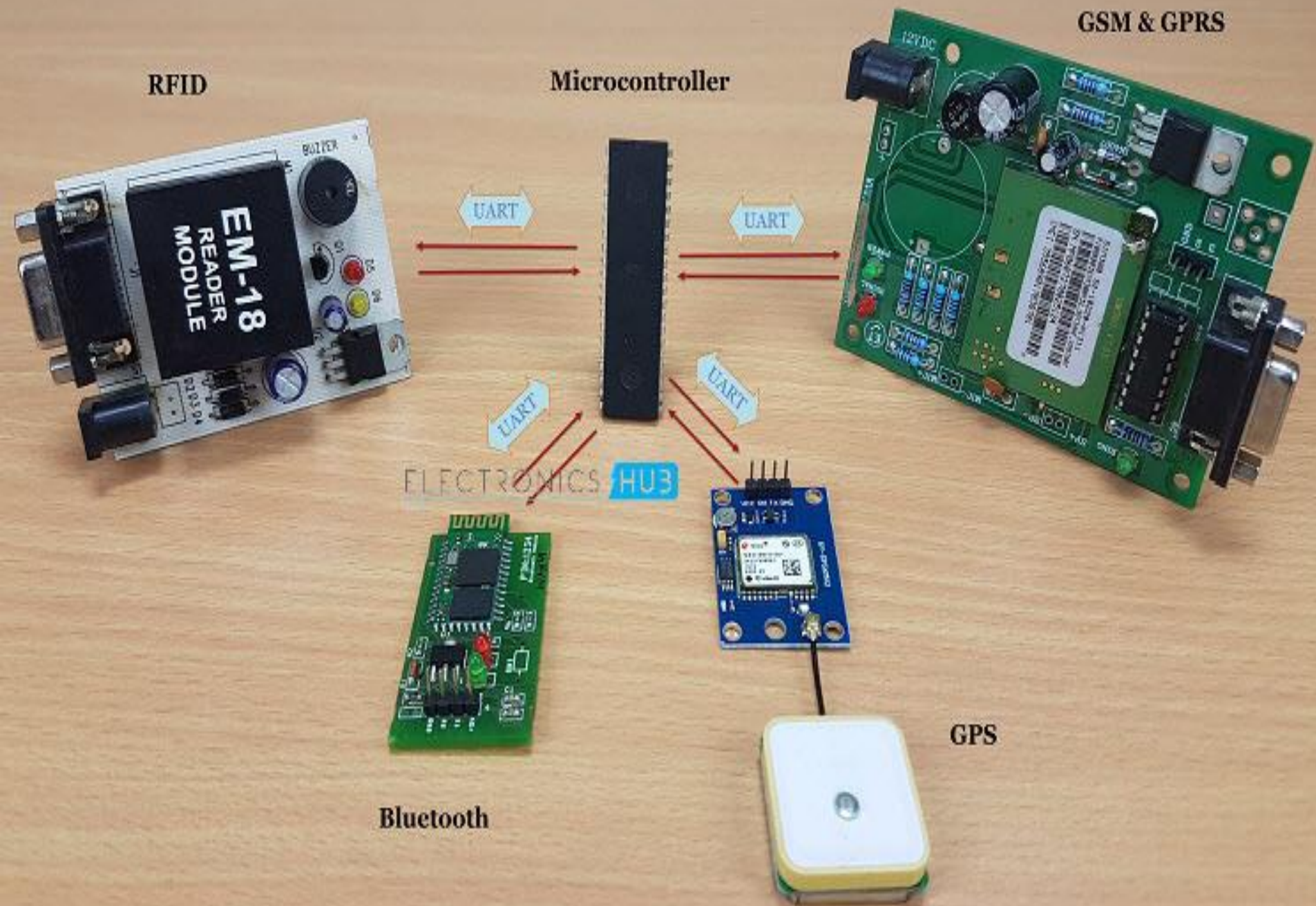
Microcontroller

GSM & GPRS

ELECTRONICS HU3

Bluetooth

GPS



RS-232
CON

SERIAL
CABLE

RS-232
CON

SUB-D 9p

CONNECT
MCU TO PC

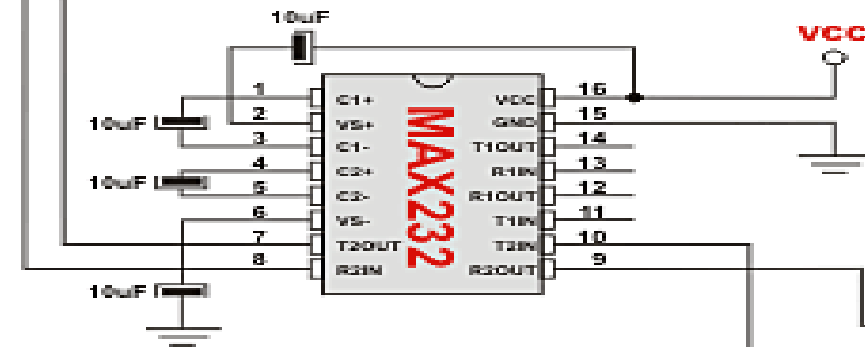
CONNECT
PC TO MCU

SUB-D 9p

Receive
data (Rx)

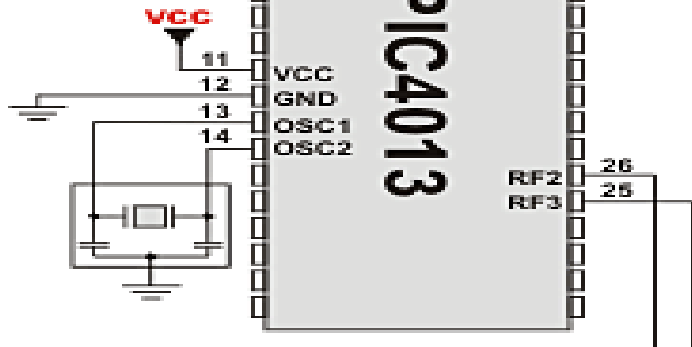
Send
Data (Tx)

PC



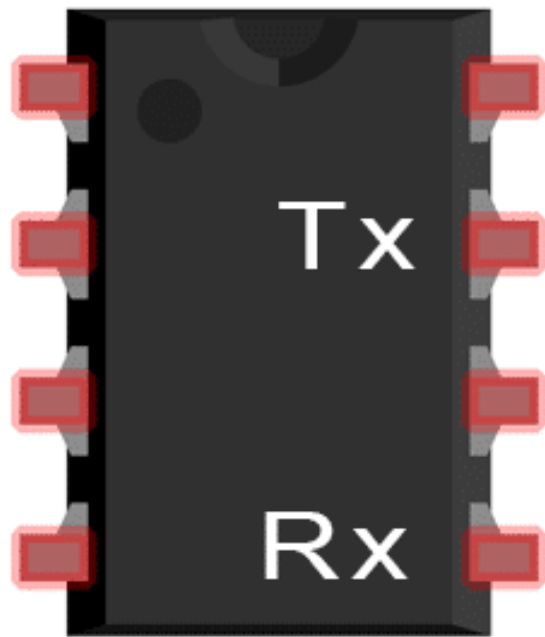
Rx

Tx

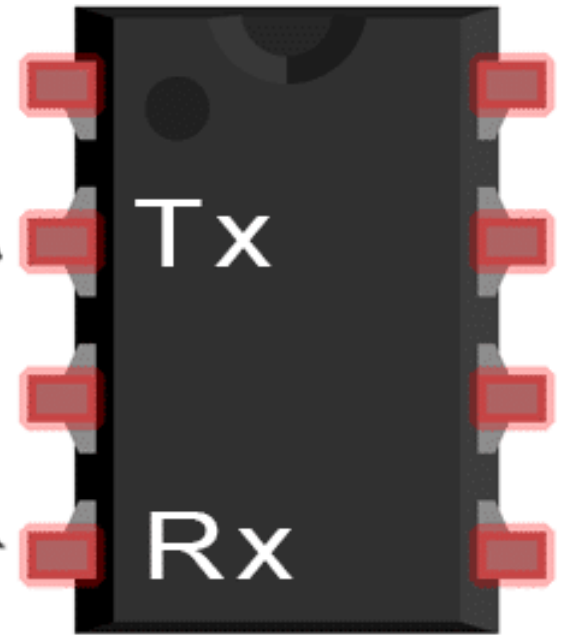


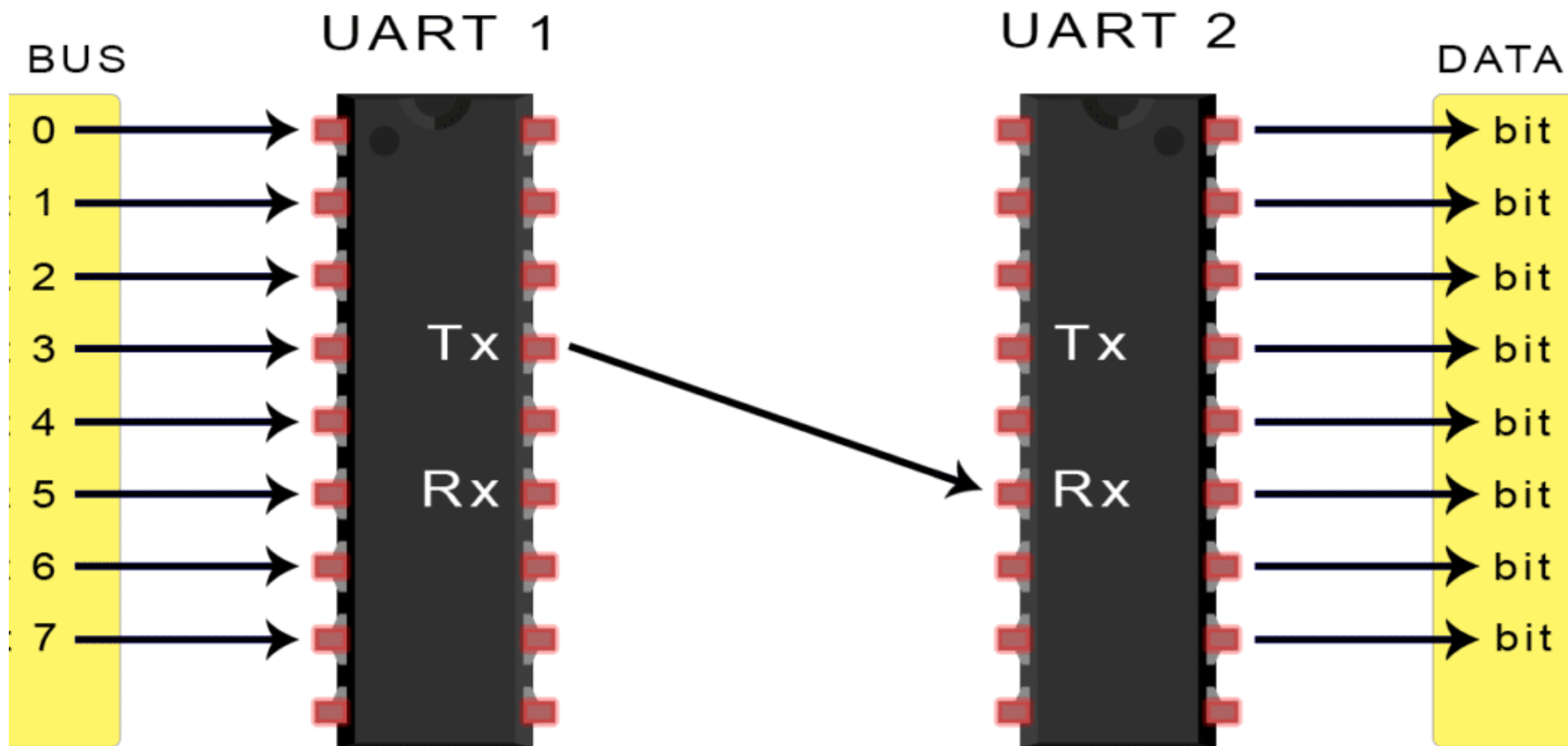
dsPIC4013

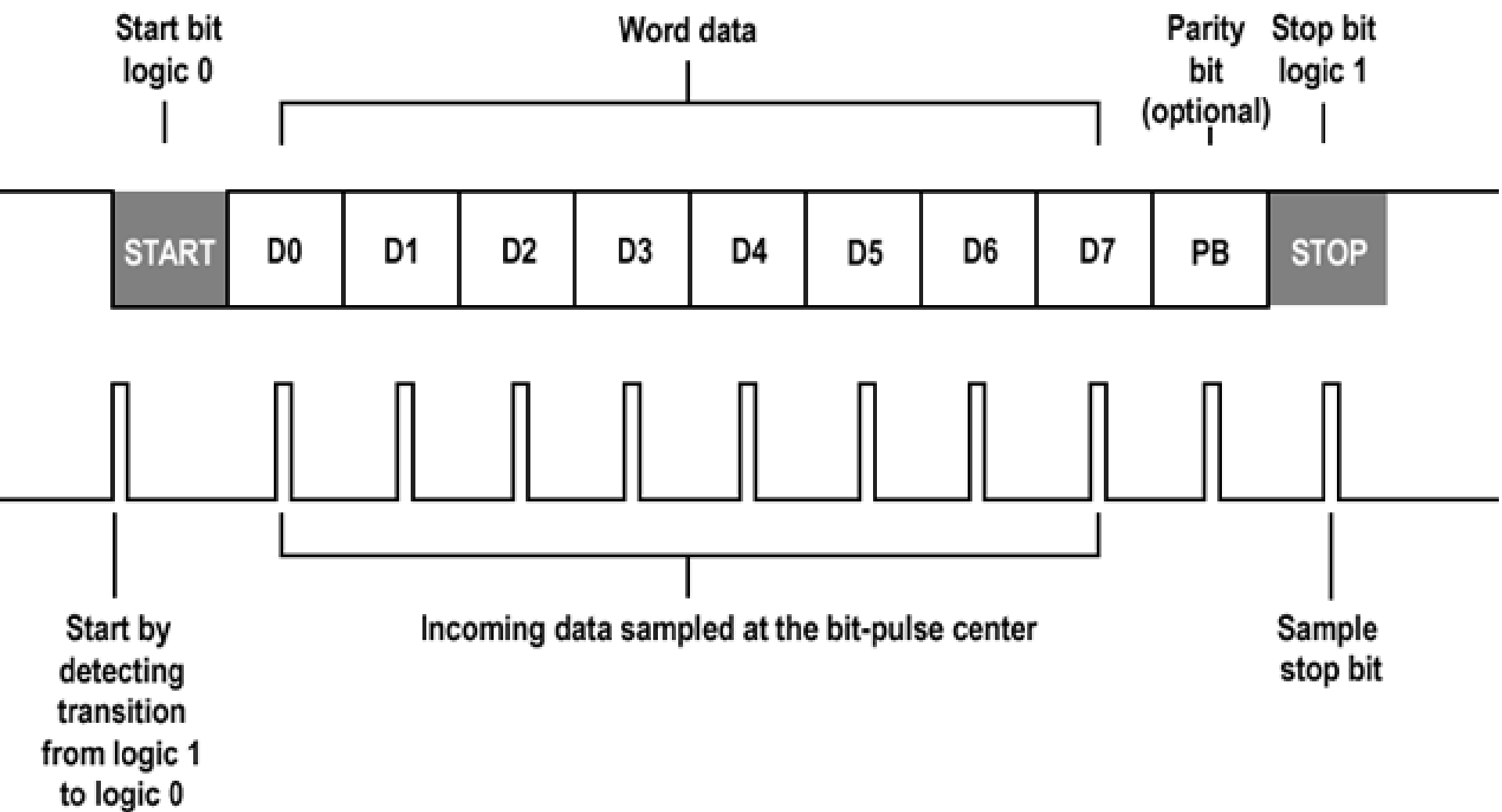
UART 1



UART 2

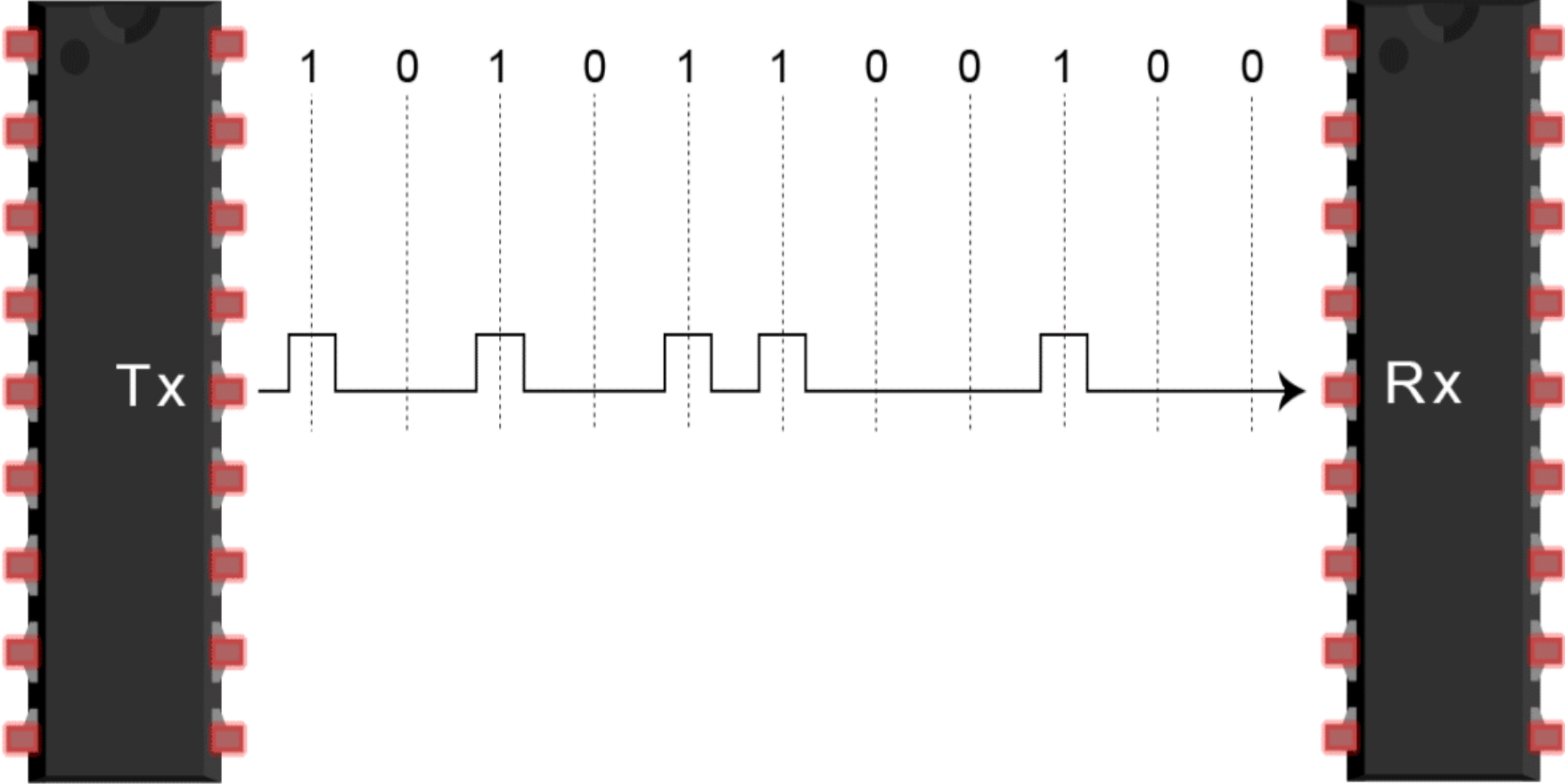


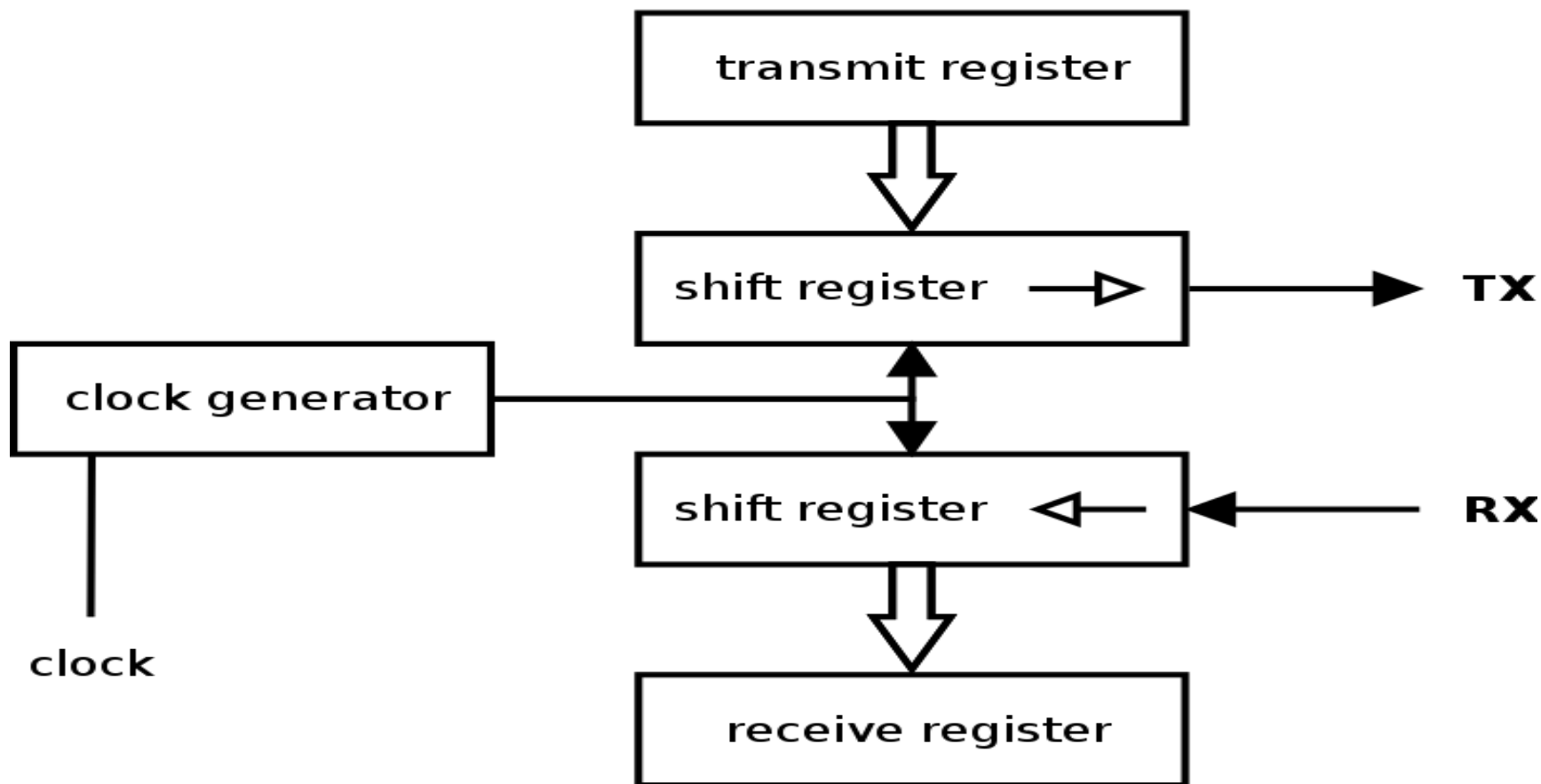


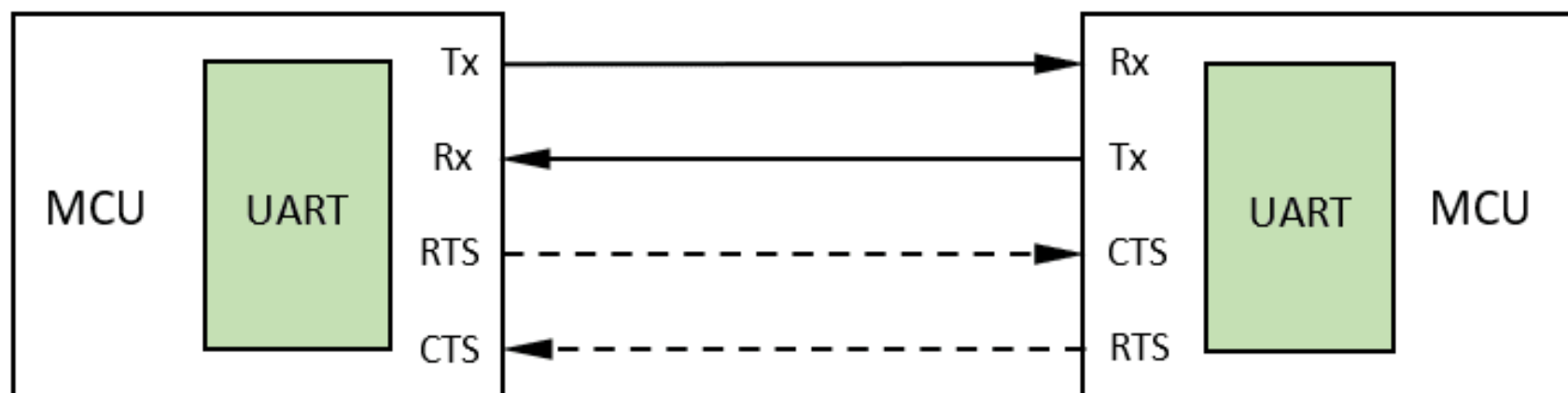


TRANSMITTING
UART

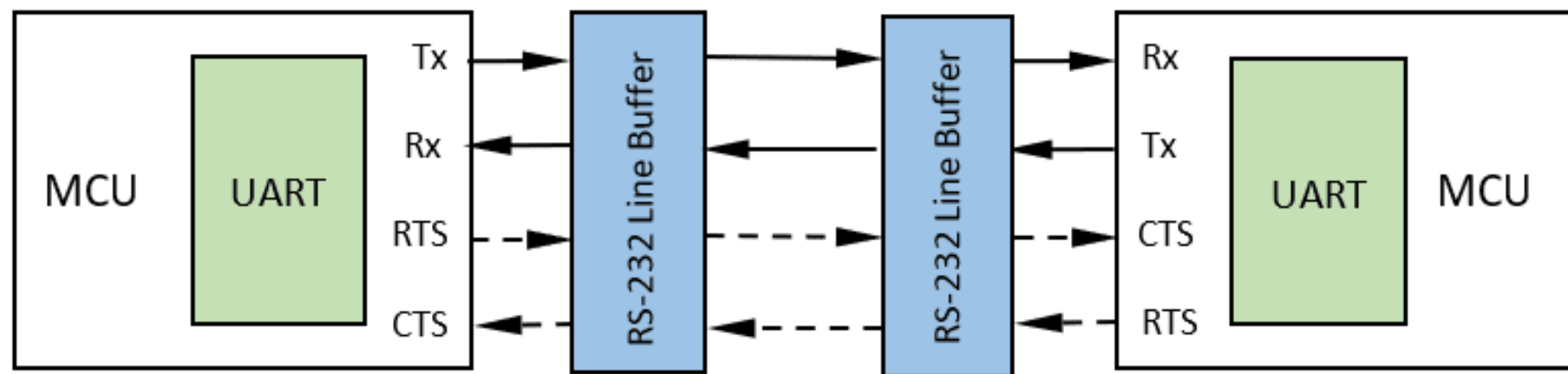
RECEIVING
UART







(a) Chip to Chip Direct UART Link

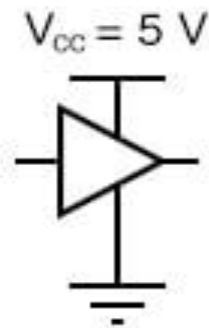
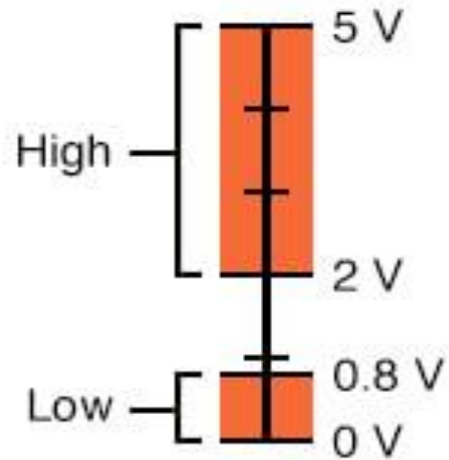


(b) Up to 5m cable RS-232 Link

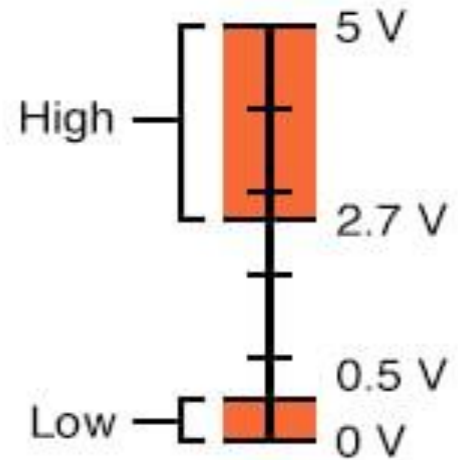
RTS = Request to Send output
CTS = Clear to Send input

Fig.2 Short and Long Serial Asynchronous Links

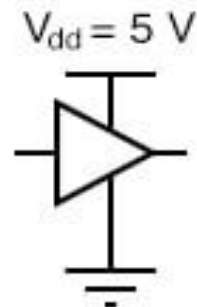
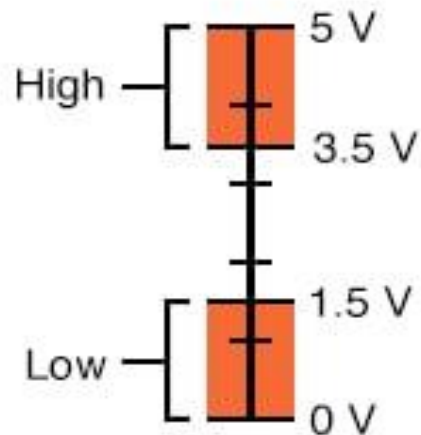
Acceptable TTL Gate
Input Signal Levels



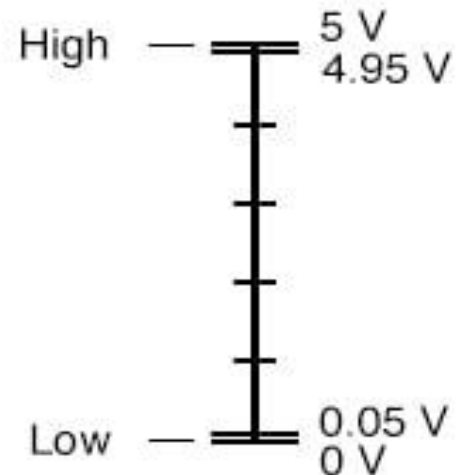
Acceptable TTL Gate
Output Signal Levels



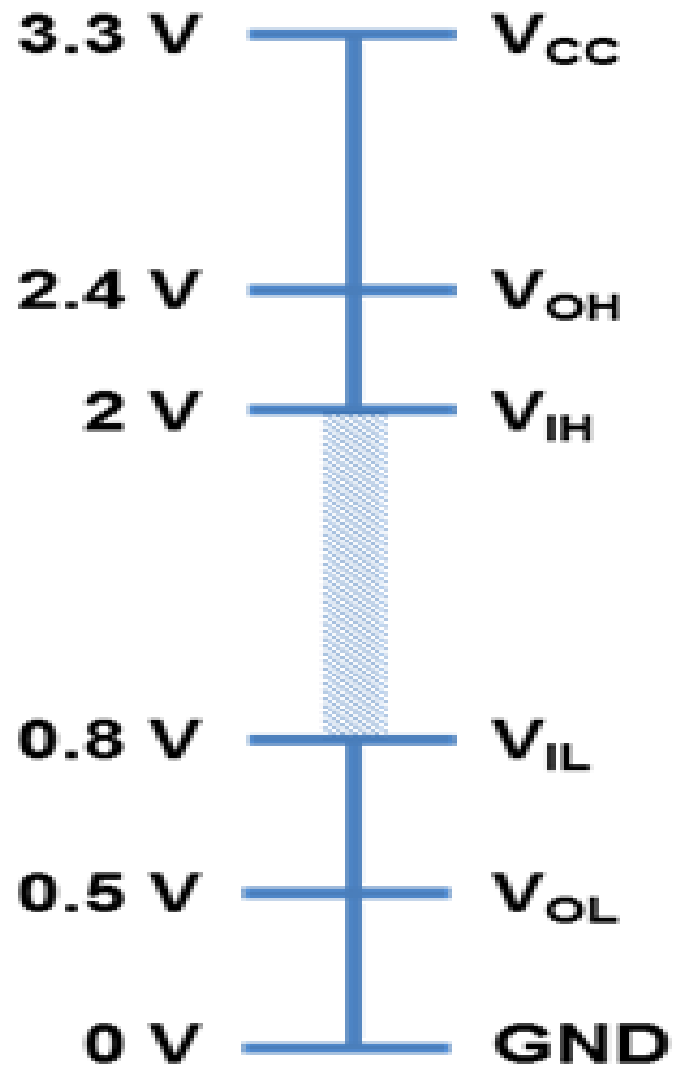
Acceptable CMOS Gate
Input Signal Levels



Acceptable CMOS Gate
Output Signal Levels



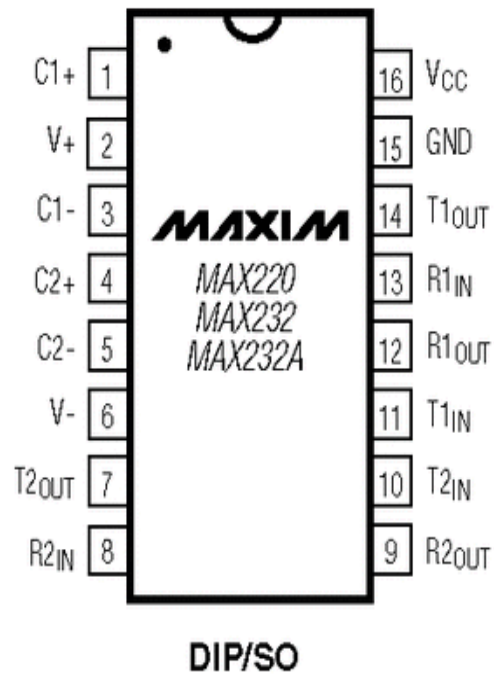
3.3 V CMOS Logic Levels



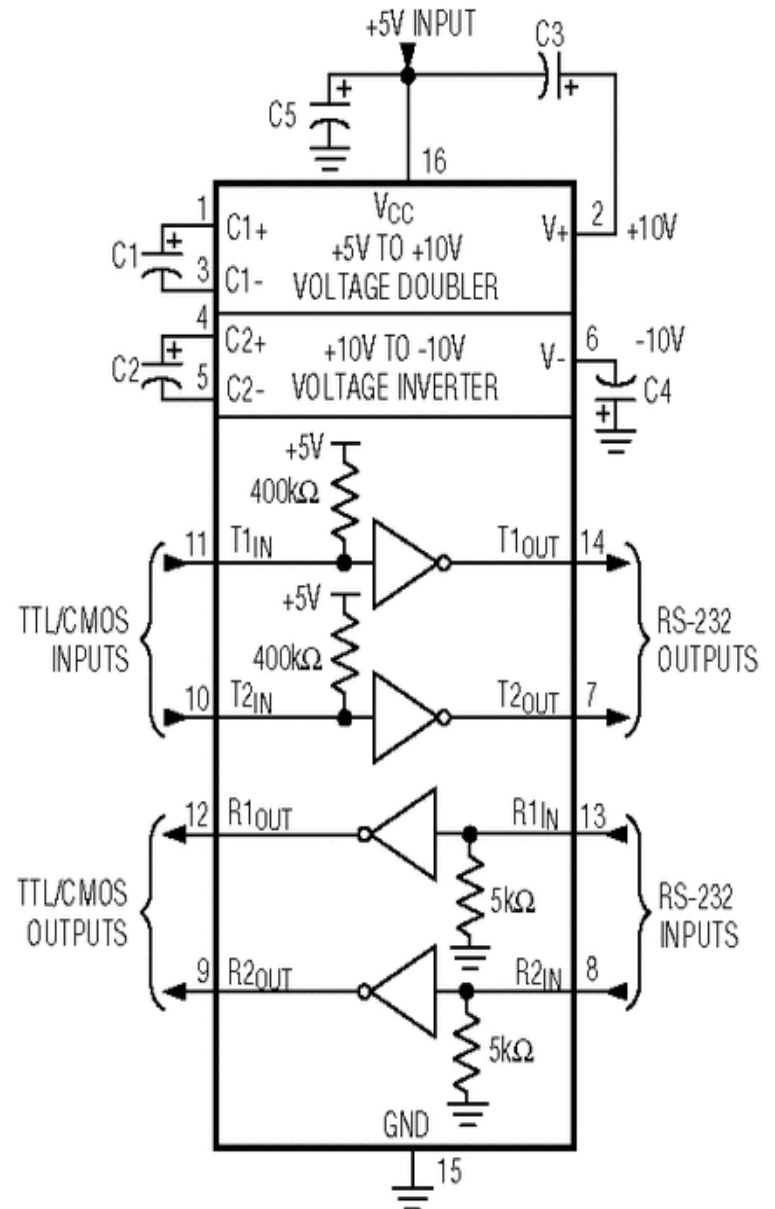
3.3 V Logic Families
(based on 74LVT04 Hex Inverter)

Source From

pilgoo.blogspot.com

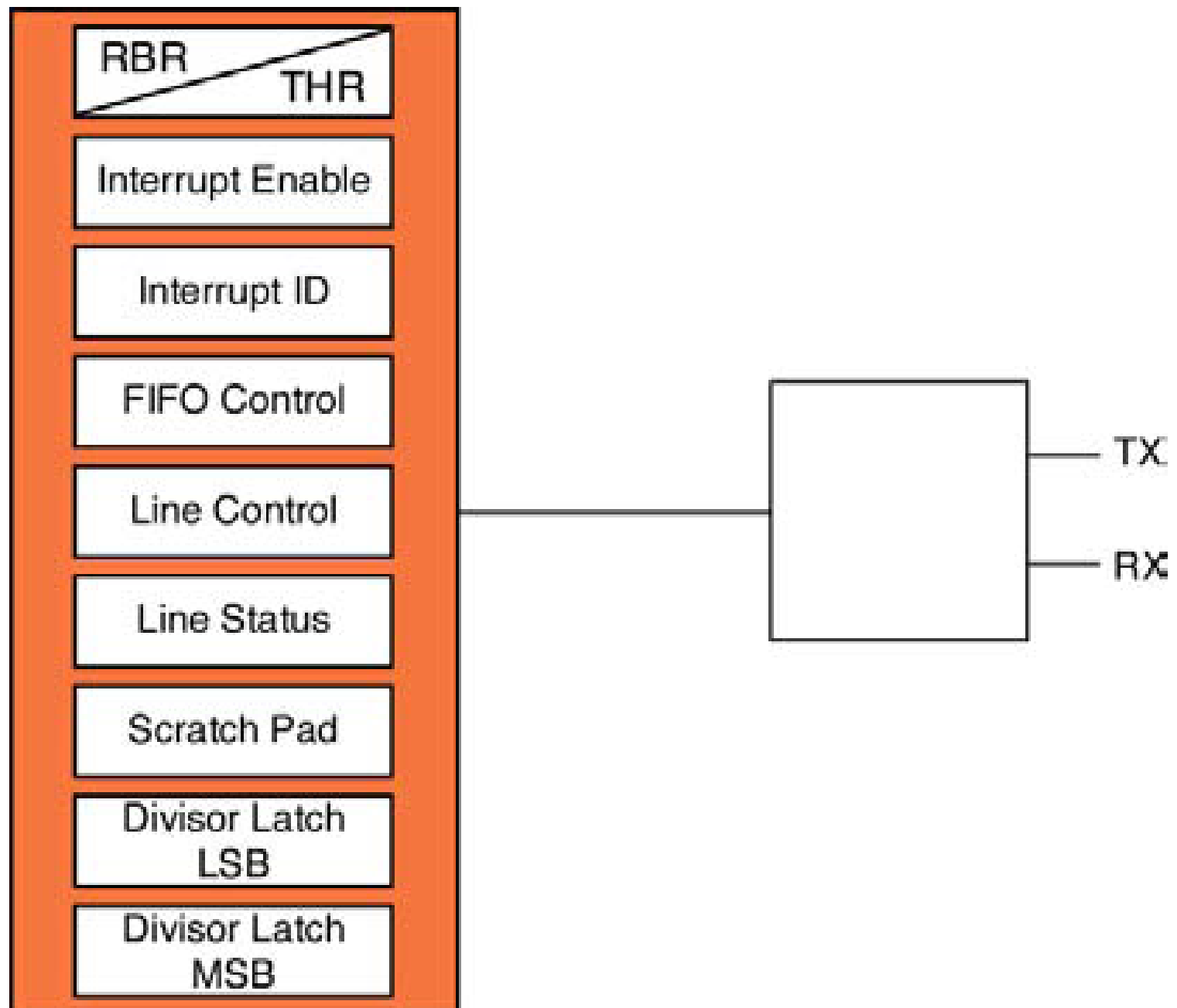


CAPACITANCE (μF)					
DEVICE	C1	C2	C3	C4	C5
MAX220	0.047	0.33	0.33	0.33	0.33
MAX232	1.0	1.0	1.0	1.0	1.0
MAX232A	0.1	0.1	0.1	0.1	0.1



UART

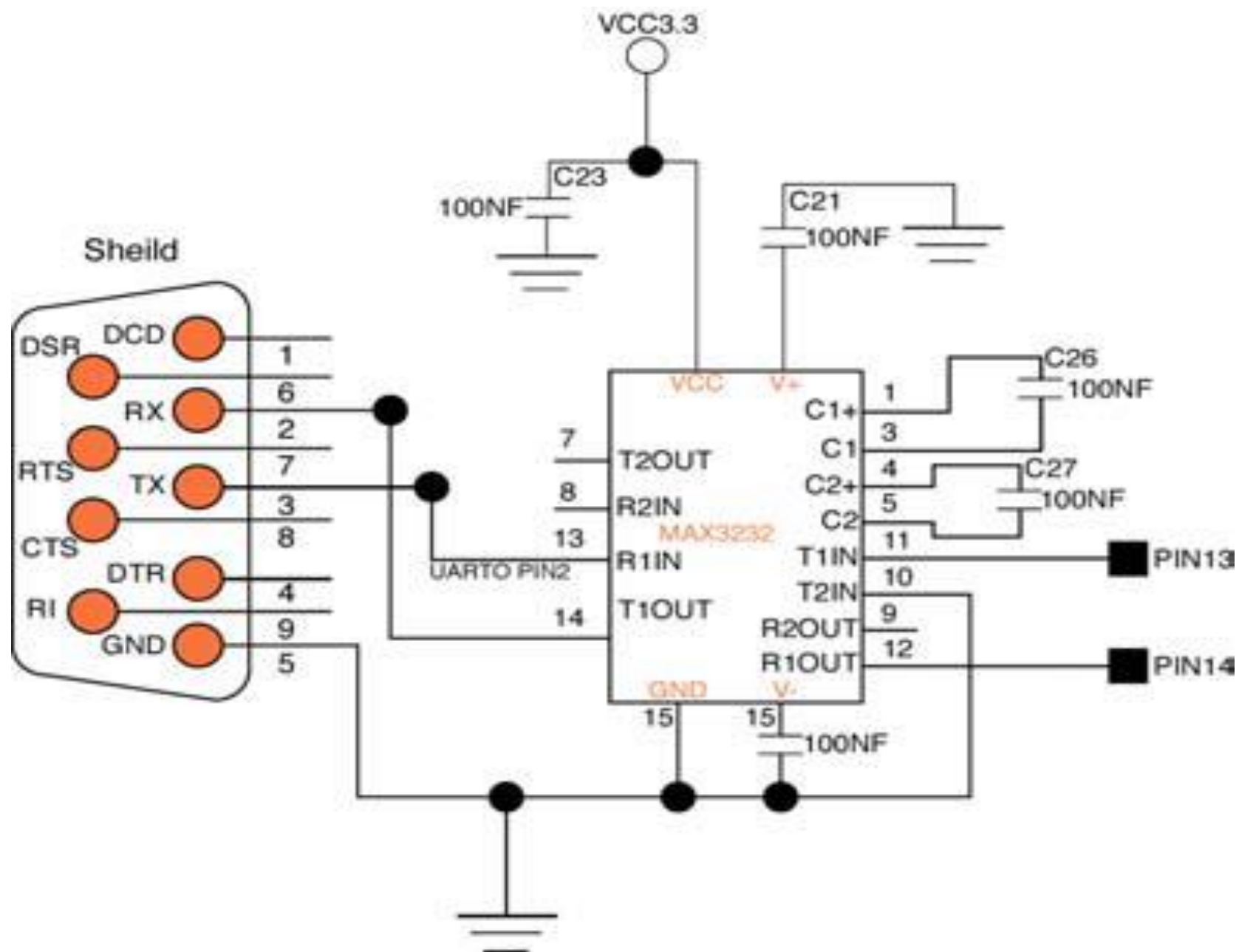
- **In ARM LPC2129, there are two UART modules named by UART0 and UART1.**
- **Considering UART0 communication, two port pins are used – P0.0(transmission/TXD0), P0.1(reception/RXD0).**
- Both have a built-in baud rate generator
- Both have 16 byte transmit and receive FIFOs.



REGISTER DESCRIPTION

Table 74: UART0 Register Map

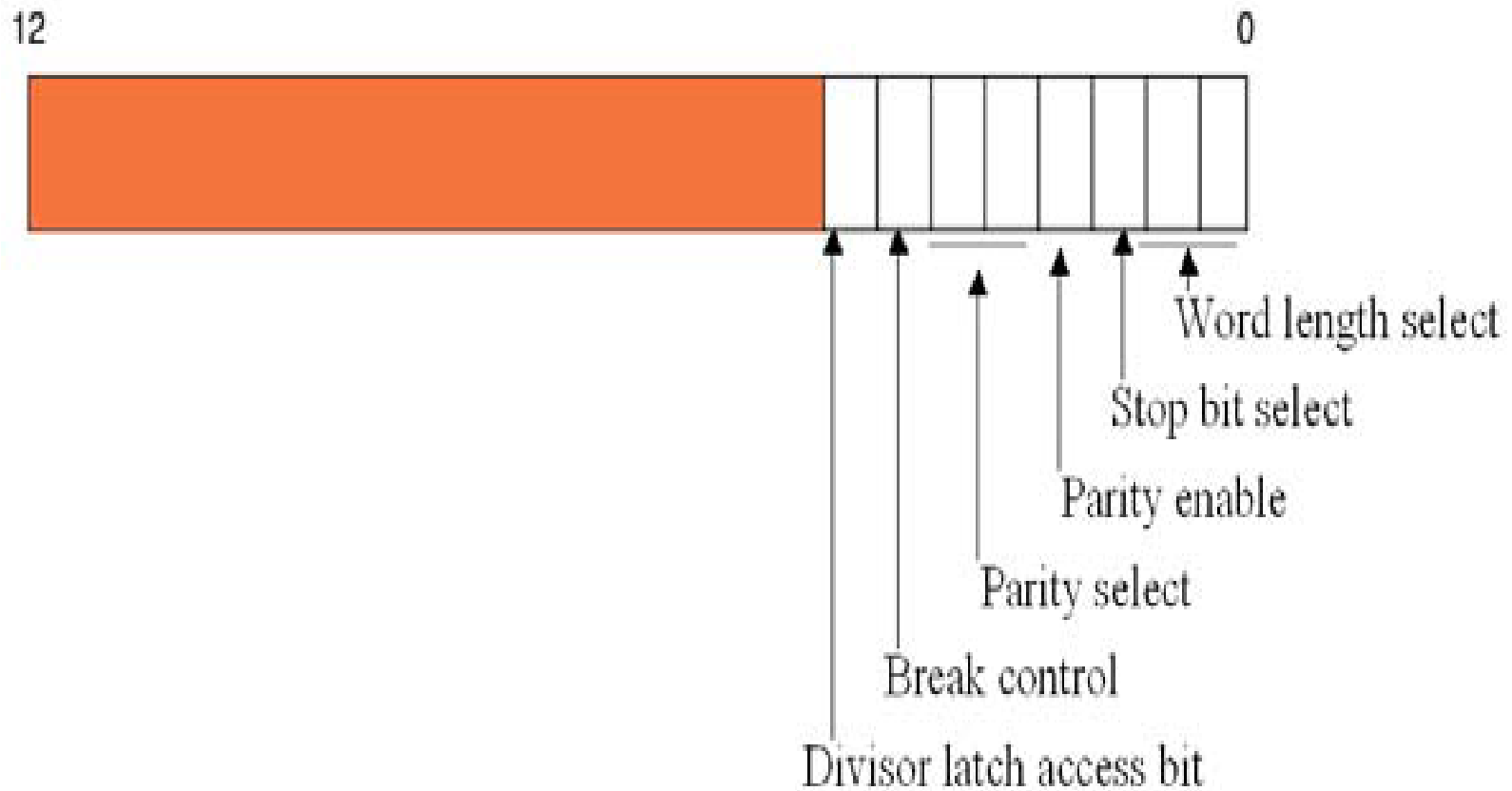
Name	Description	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0	Access	Reset Value*	Address
U0RBR	Receiver Buffer Register	MSB READ DATA LSB								RO	un-defined	0xE000C000 DLAB = 0
U0THR	Transmit Holding Register	MSB WRITE DATA LSB								WO	NA	0xE000C000 DLAB = 0
U0IER	Interrupt Enable Register	0	0	0	0	0	Enable Rx Line Status Interrupt	Enable THRE Interrupt	Enable Rx Data Available Interrupt	R/W	0	0xE000C004 DLAB = 0
U0IIR	Interrupt ID Register	FIFOs Enabled		0	0	IIR3	IIR2	IIR1	IIR0	RO	0x01	0xE000C008
U0FCR	FIFO Control Register	Rx Trigger		Reserved		-	Tx FIFO Reset	Rx FIFO Reset	FIFO Enable	WO	0	0xE000C008
U0LCR	Line Control Register	DLAB	Set Break	Stick Parity	Even Parity Select	Parity Enable	Number of Stop Bits	Word Length Select		R/W	0	0xE000C00C
U0LSR	Line Status Register	Rx FIFO Error	TEMT	THRE	BI	FE	PE	OE	DR	RO	0x60	0xE000C014
U0SCR	Scratch Pad Register	MSB LSB								R/W	0	0xE000C01C
U0DLL	Divisor Latch LSB	MSB LSB								R/W	0x01	0xE000C000 DLAB = 1
U0DLM	Divisor Latch MSB	MSB LSB								R/W	0	0xE000C004 DLAB = 1



Initialization of the UART0

- Program
- pinselect block to switch the processor pins from GPIO to the UART functions.
- UART line control register (U0LCR) is used to configure the format of the transmitted data character.
- *Setting the DLAB bit allows programming of the BAUD rate generators*

LCR



- To calculate the UART baud rate for the baud rate generator is:

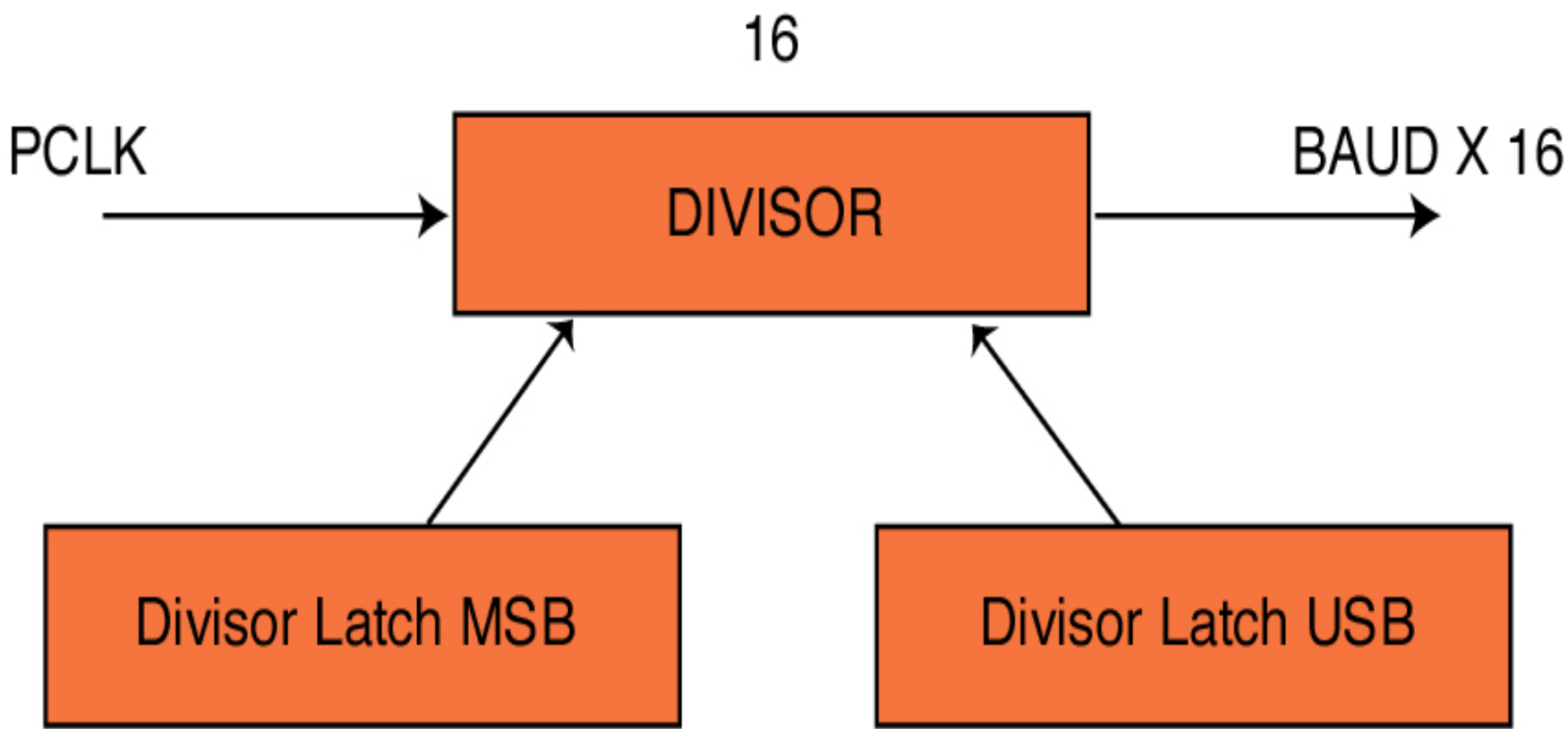
$$PCLK = CCLK \text{ or } CCLK/2 \text{ or } CCLK/4$$

$$\text{Divisor} = PCLK / (16 \times \text{BAUD})$$

- Hence

$$U0DLL = \text{Divisor}$$

$$U0DLM = \text{Divisor} \gg 8$$



DLAB = 1

- LPC2129 UARTs will work with up to around a 5% error in the bit timing.
- The divisor value is held in two registers, Divisor latch MSB (DLM) and Divisor latch LSB (DLL).
- The first eight bits of both registers holds each half of the divisor as shown below.
- Finally the DLAB bit in the LCR register must be set back to zero to protect the contents of the divisor registers.

Once the UART is initialized,

- Characters can be transmitted by writing to the **Transmit Holding Register**.
- Characters may be received by reading from the **Receive Buffer Register**.
- In fact both these registers occupy the same memory location, writing a character places the character in the transmit FIFO
- and reading from this location loads a character from the Receive FIFO.

- The **Line Status Register(LSR)** is used to check on UART error conditions and to check the status of the receive and transmit FIFOs.
- *The LSR contains flags which indicate events within the UART. It may be polled or should be read after a UART interrupt is generated.*

Line Status Register (LSR)

